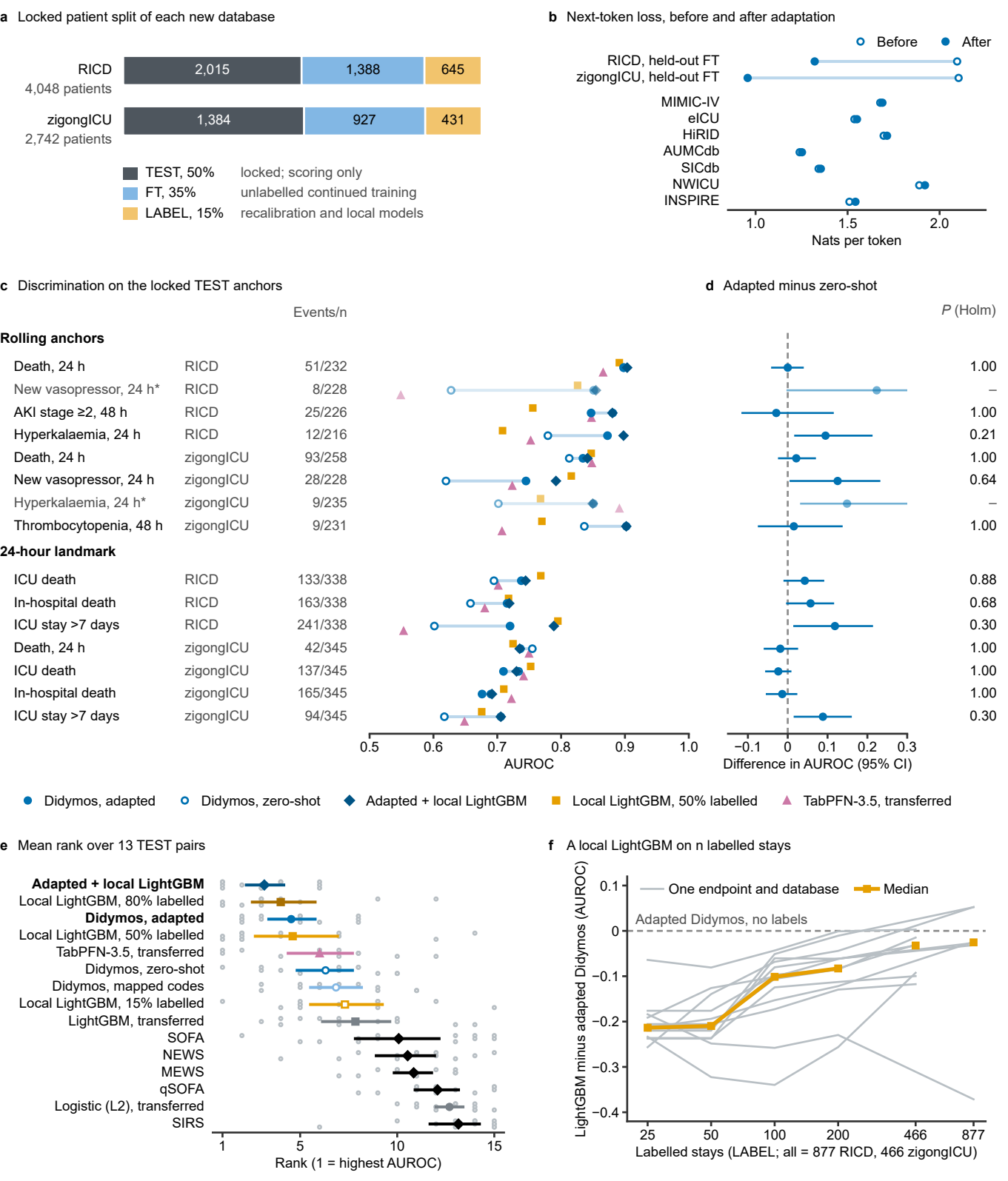

a Locked patient split of each new database
RICD
4,048 patients
2,015
1,388
645
zigongICU
2,742 patients
1,384
927
431
TEST, 50%
locked; scoring only
FT, 35%
unlabelled continued training
LABEL, 15%
recalibration and local models
b Next-token loss, before and after adaptation
Before
After
RICD, held-out FT
zigongICU, held-out FT
MIMIC-IV
eICU
HiRID
AUMCdb
SICdb
NWICU
INSPIRE
1.0
1.5
2.0
Nats per token
c Discrimination on the locked TEST anchors
Events/n
Rolling anchors
Death, 24 h RICD 51/232
New vasopressor, 24 h* RICD 8/228
AKI stage ≥2, 48 h RICD 25/226
Hyperkalaemia, 24 h RICD 12/216
Death, 24 h zigongICU 93/258
New vasopressor, 24 h zigongICU 28/228
Hyperkalaemia, 24 h* zigongICU 9/235
Thrombocytopenia, 48 h zigongICU 9/231
24-hour landmark
ICU death RICD 133/338
In-hospital death RICD 163/338
ICU stay >7 days RICD 241/338
Death, 24 h zigongICU 42/345
ICU death zigongICU 137/345
In-hospital death zigongICU 165/345
ICU stay >7 days zigongICU 94/345
0.5
0.6
0.7
0.8
0.9
1.0
AUROC
d Adapted minus zero-shot
P (Holm)
1.00
–
1.00
0.21
1.00
0.64
–
1.00
0.88
0.68
0.30
1.00
1.00
1.00
0.30
−0.1
0
0.1
0.2
0.3
Difference in AUROC (95% CI)
Didymos, adapted
Didymos, zero-shot
Adapted + local LightGBM
Local LightGBM, 50% labelled
TabPFN-3.5, transferred
e Mean rank over 13 TEST pairs
Adapted + local LightGBM
Local LightGBM, 80% labelled
Didymos, adapted
Local LightGBM, 50% labelled
TabPFN-3.5, transferred
Didymos, zero-shot
Didymos, mapped codes
Local LightGBM, 15% labelled
LightGBM, transferred
SOFA
NEWS
MEWS
qSOFA
Logistic (L2), transferred
SIRS
1
5
10
15
Rank (1 = highest AUROC)
f A local LightGBM on n labelled stays
One endpoint and database
Median
Adapted Didymos, no labels
LightGBM minus adapted Didymos (AUROC)
0.1
0
−0.1
−0.2
−0.3
−0.4
25
50
100
200
466
877
Labelled stays (LABEL; all = 877 RICD, 466 zigongICU)